Decision Updating in Large Language Models
Contradictory Evidence §2; Appendix C
Sources and conditions §2.1; C.1
Direct contradiction / corrective evidence
Belief-R: revision and preservation | Wilie et al. (2024)
Context-memory disagreement | Xie et al. (2024a)
Temporal change / stale information
SituatedQA: time-dependent answers | Zhang and Choi (2021)
Source disagreement
WikiContradict: conflicting passages | Hou et al. (2024)
Misleading or invalid counterevidence
ClashEval: perturbed evidence | Wu et al. (2024)
FlipFlop: unsupported challenges | Laban et al. (2023)
Model behaviours §2.2; C.2
Appropriate revision / justified resistance
Revision-preservation trade-off | Wilie et al. (2024)
Anchoring / overcorrection
Confirmation bias | Xie et al. (2024a)
Overriding correct prior answers | Wu et al. (2024)
Sycophancy / answer reversal
Agreement with user beliefs | Sharma et al. (2024)
Accuracy loss after challenges | Laban et al. (2023)
Order and source effects / confidence changes
Evidence selection and presentation | Xie et al. (2024a)
Confidence-dependent context adoption | Wu et al. (2024)
Evaluation §2.3; C.3
Conflict handling / source arbitration
WikiContradict | Hou et al. (2024)
ClashEval | Wu et al. (2024)
Revision / preservation / correction selectivity
Belief-R | Wilie et al. (2024)
Supporting, unrelated, and exceptional evidence | Leidinger et al. (2024)
Propagation to dependent conclusions
MQuAKE: multi-hop consequences of edits | Zhong et al. (2023)
Mitigation §2.3; C.4
Evidence verification
Independent checks
POST
Chain-of-Verification | Dhuliawala et al. (2024)
Retrieval filtering
Relevance and critique
PRE
POST
Self-RAG | Asai et al. (2024)
Source consolidation
Reliability assessment
POST
Astute RAG | Wang et al. (2025)
Response control
Context-aware decoding
POST
CAD: increase context faithfulness | Shi et al. (2024)
Confidence-guided selection
POST
Confidence-based arbitration | Wu et al. (2024)
Training
Anti-sycophancy
PRE
Synthetic-data intervention | Wei et al. (2023)
Fine-tuning against answer flipping | Laban et al. (2023)
Complicating Evidence §3; Appendix D
Sources and conditions §3.1; D.1
New constraints / resource limitations
TravelPlanner: budgets and feasibility | Xie et al. (2024b)
FollowBench: accumulated requirements | Jiang et al. (2024)
Preferences / competing objectives
Flex-TravelPlanner: prioritized constraints | Oh et al. (2025)
Context changes / sequential requirements
Multi-IF: multi-turn instructions | He et al. (2024)
Progressively specified tasks | Laban et al. (2025)
Scope qualifications / exceptions
Defeasible NLI | Rudinger et al. (2020)
Generics and exceptions | Leidinger et al. (2024)
Contraindications / policy restrictions
Domain policies limiting permitted actions | Yao et al. (2024)
Model behaviours §3.2; D.2
Appropriate modification / constraint neglect
Adaptation to new requirements | Oh et al. (2025)
Tracking multiple constraints | Xie et al. (2024b)
Unnecessary reversal / loss of valid commitments
Instability under irrelevant additions | Leidinger et al. (2024)
New preferences displacing prior constraints | Oh et al. (2025)
Premature commitment / overgeneralisation
Early assumptions and failed recovery | Laban et al. (2025)
Generalizations and exceptions | Leidinger et al. (2024)
Coexisting requirements / priority handling
Satisfying multiple constraints | Jiang et al. (2024)
Explicitly prioritized requirements | Oh et al. (2025)
Evaluation §3.3; D.3
Conditional reasoning / constraint satisfaction
Defeasible NLI | Rudinger et al. (2020)
IFEval: executable instruction checks | Zhou et al. (2023)
Preference prioritization / dynamic planning
Sequential and priority-based settings | Oh et al. (2025)
Longitudinal retention / updated information
Multi-IF | He et al. (2024)
LongMemEval: memory and knowledge updates | Wu et al. (2025)
Mitigation §3.3; D.4
Explicit representations
Constraint tracking
POST
LLM+P: PDDL problem specification | Liu et al. (2023)
Task organization
Decomposition
POST
Plan-and-Solve prompting | Wang et al. (2023)
External support
Classical solvers
POST
LLM+P: planning with external solvers | Liu et al. (2023)
Verification
POST
LLM-Modulo: external critics | Kambhampati et al. (2024)
Memory support
Context management
POST
MemGPT: hierarchical memory | Packer et al. (2023)
Feedback and adaptation
Replanning / refinement
POST
ReAct: update action plans | Yao et al. (2023)
Self-Refine: iterative feedback | Madaan et al. (2023)
Uncertainty-Inducing Evidence §4; Appendix E
Sources and conditions §4.1; E.1
Missing information / contextual insufficiency
Sufficient Context | Joren et al. (2024)
MediQ: incomplete clinical information | Li et al. (2024)
Ambiguity / competing interpretations
AmbigQA | Min et al. (2020)
CLAMBER | Zhang et al. (2024)
Unreliable evidence / unresolved explanations
Unresolved passage disagreement | Hou et al. (2024)
Incorrect retrieved evidence | Wu et al. (2024)
Outcome uncertainty / high-risk settings
KnowNo: uncertain action selection | Ren et al. (2023)
Clinical information seeking | Li et al. (2024)
Model behaviours §4.2; E.2
Uncertainty recognition / confidence adjustment
Ambiguity identification | Zhang et al. (2024)
Overconfidence and elicitation | Xiong et al. (2024)
Information seeking / conditional responses
CLAM: clarification before answering | Kuhn et al. (2022)
Follow-up questions before commitment | Li et al. (2024)
Premature closure / false certainty
Premature solutions | Laban et al. (2025)
Answering despite insufficient context | Joren et al. (2024)
Abstention / assistance and escalation
AbstentionBench | Kirichenko et al. (2025)
KnowNo: asking for human help | Ren et al. (2023)
Excessive caution / unnecessary refusal
XSTest: exaggerated safety behaviour | Röttger et al. (2024)
Evaluation §4.3; E.3
Calibration / failure prediction
Confidence calibration and discrimination | Xiong et al. (2024)
User-facing linguistic calibration | Band et al. (2024)
Uncertainty estimation / selective prediction
Semantic entropy | Kuhn et al. (2023)
Selective QA: accuracy-coverage trade-off | Kamath et al. (2020)
Clarification quality / ambiguity recognition
CLAMBER | Zhang et al. (2024)
CLAM: interactive clarification | Kuhn et al. (2022)
Decisions under uncertainty / assistance seeking
KnowNo: task completion and human help | Ren et al. (2023)
Mitigation §4.3; E.4
Confidence and expression
Elicitation / prompting
POST
Prompting, sampling, and aggregation | Xiong et al. (2024)
Calibration training
PRE
Verbalized probability training | Lin et al. (2022)
Linguistic calibration | Band et al. (2024)
Evidence acquisition
Selective retrieval
PRE
POST
Self-RAG: adaptive retrieval and critique | Asai et al. (2024)
Clarification
POST
CLAM: detect, clarify, answer | Kuhn et al. (2022)
MediQ: adaptive question asking | Li et al. (2024)
Response selection
Selective generation
PRE
POST
Context-guided abstention | Joren et al. (2024)
Trained error predictor and abstention | Kamath et al. (2020)
Assistance seeking
PRE
POST
KnowNo: conformal prediction and help | Ren et al. (2023)
Cross-Cutting Synthesis (Orthogonal) §5; A.4–A.5; Appendix F
Evidence properties §5; A.5
Conflict location
Context-memory; inter-context; intra-memory | Xu et al. (2024)
Reliability / provenance / evidence dependence
Source-aware reliability | Wang et al. (2025)
Agreement and evidence presentation | Xie et al. (2024a)
Temporal validity
Time and geographical applicability | Zhang and Choi (2021)
Temporal reasoning and knowledge updates | Wu et al. (2025)
Decision and interaction §5; A.4–A.5
Decision scope / affected components
Dependent conclusions | Zhong et al. (2023)
Actions and retained constraints | Oh et al. (2025)
Response distinctions
Revision versus preservation | Wilie et al. (2024)
Strengthening or weakening an inference | Rudinger et al. (2020)
Interaction order / length and continuity
Position of relevant information | Liu et al. (2024)
Multi-session memory | Wu et al. (2025)
Rationale faithfulness / expressed confidence
Explanations can omit causal influences | Turpin et al. (2023)
How statements inform user judgments | Band et al. (2024)
Settings and interpretation §5; B.3; F.4–F.6
Applications / high-stakes decisions
Travel planning | Xie et al. (2024b)
Clinical reasoning | Li et al. (2024)
Embodied assistance | Ren et al. (2023)
Language / modality
Multi-IF: eight languages | He et al. (2024)
Embodied planning; not universal multimodal coverage | Ren et al. (2023)
Metrics / prompts / evaluation robustness
Prompt-format sensitivity | Sclar et al. (2024)
LLM-judge biases | Zheng et al. (2023)

§ = main-text section; lettered references = appendix subsections
PRE = preparatory training / calibration; POST = evidence-time inference / response processing; both = combined stages.